

Markov Decision Processes

Chapter 3 of Sutton Barto

- A process that captures the interaction between the agent and environment
- - States belong to environment
 - Action - - - agent
 - Reward - - - environment
- Interaction is captured via the following sequence:
$$\underbrace{s_0, A_0, R_1, s_1}_{\text{}} \underbrace{A_1, R_2, s_2, \dots}_{\text{}}$$
- Assume the system model is known

- Assume that

$$P(s', r | s, a)$$

$$\triangleq P(s_{t+1} = s', R_{t+1} = r | s_t = s, A_t = a)$$

are known.

- The above is a general model
- More specialized model.

$$P(s' | s, a)$$

$$= P(s_{t+1} = s' | s_t = s, A_t = a)$$

$$= \sum_{r \in \mathcal{R}} P(s', r | s, a)$$

- Expected single stage reward as a 2-argument function

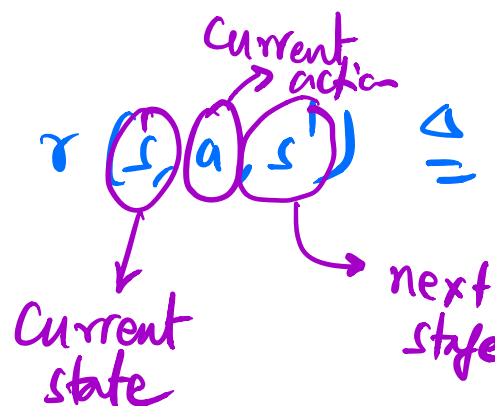
$$r: S \times A \rightarrow \mathbb{R}$$

$$r(s, a) \triangleq E \left[R_{t+1} \mid s_t = s, A_t = a \right]$$

$$= \sum_{r \in \mathbb{R}} r \sum_{s' \in S} P(s', r \mid s, a)$$

- Expected single stage reward as a 3-argument function

$$r: S \times A \times S \rightarrow \mathbb{R}$$

$$r(s, a, s') \triangleq E \left[R_{t+1} \mid s_t = s, A_t = a, s_{t+1} = s' \right]$$


Example: Recycling Robot

Setting:

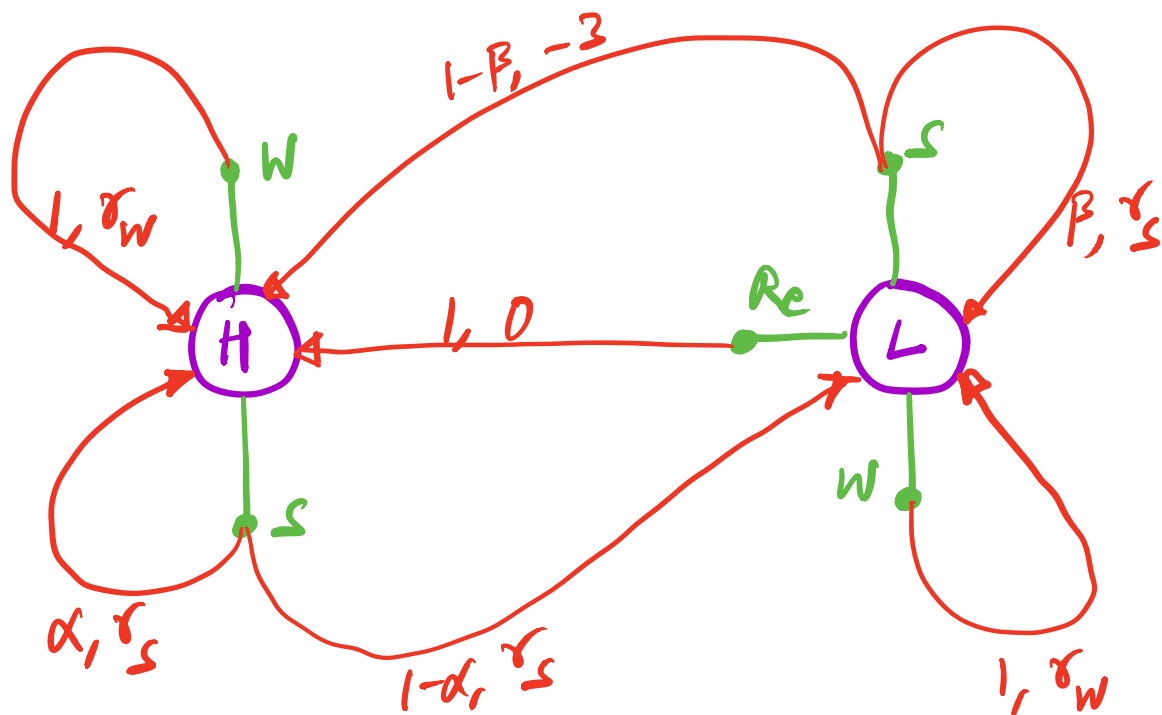
- A mobile robot deployed in ~~office~~ office
- Robot collects softdrink cans
 - Move around & collect cans
 - wait in a corner for people to bring cans
- Robot's decisions will be influenced by it's charge level
- State = charge level of the robot

- state = high (H) or low (L)
- Actions
 - Actions in H = Search (s)
or
wait (w)
 - Actions in L = s, w,
Recharge (Re)
- Rewards
 - > 0 if Robot
secures an
empty can
 - < 0 if battery
charge
becomes
0

Consider the following transition table

State	Action	Next state	transition prob.	Reward
H	S	H	α	r_s
H	S	L	$1-\alpha$	r_s
L	S	L	β	r_s
L	S	H	$1-\beta$	-3
L	W	L	1	r_w
L	Re	H	1	0
H	W	H	1	r_w

Alternatively, we may represent the above via a diagram



State transition diagram

One may let $r_s > r_w$

- Reward is a design objective

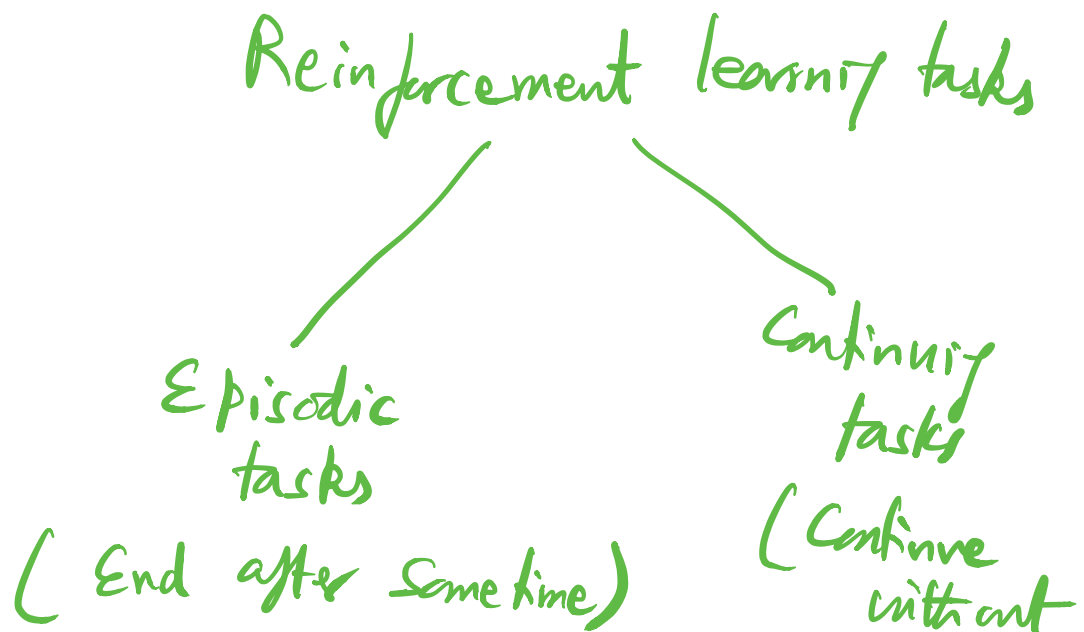
Let $R_{t+1}, R_{t+2}, \dots$ be a sequence

of rewards obtained after time t

- Let $G_t = R_{t+1} + R_{t+2} + \dots + R_T$

Here $T \equiv$ final time step

$G_t \equiv$ Return on a trajectory
starting from time t .



stopping)

Episodes: Trajectories of states,
actions, rewards and next states
that end in a final state S_T
goal state

and a final reward R_T at termination
time T .

- When an episode terminates,
we start with a new episode
from either same initial state as
previous episode or by picking
an initial state randomly.

- Let $S \equiv$ set of nonterminal states

$S_T \equiv$ terminal state

- Let $S^+ = S \cup \{S_T\}$

$\equiv$ set of all states

- For continuing tasks, we have
discounted rewards

Let $0 \leq \gamma < 1$ be the

discount factor.

Returns

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} \\ + \dots + \gamma^k R_{t+k+1} + \dots$$

- Effect of γ :

- γ is close to 0

→ Agent puts max emphasis on immediate rewards

- γ close to 1

→ Agent puts more emphasis on rewards that come in future

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

$$= R_{t+1} + \gamma (R_{t+2} + \gamma R_{t+3} + \dots)$$

$$= R_{t+1} + \gamma G_{t+1}$$

- The above holds even in terminal tasks if $G_T = 0$
- We let $\gamma < 1$ in continuing tasks because if $|R_t| \leq B < \infty$
 $\forall t$

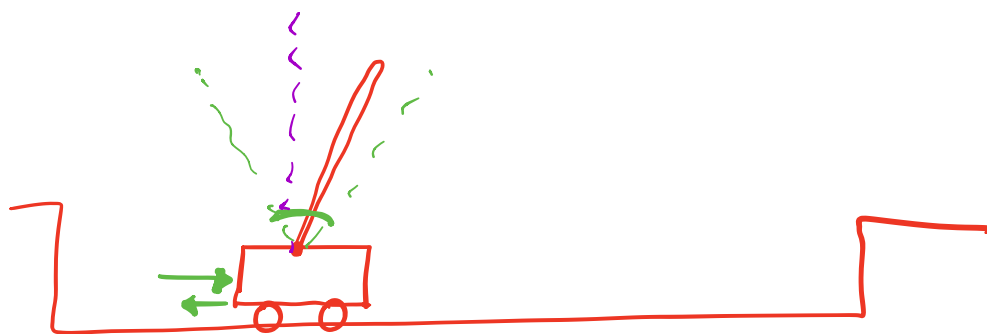
then

$$|G_t| \leq \sum_{k=0}^{\infty} \gamma^k B$$

$$= \frac{B}{(1-\gamma)} < \infty$$

if
 $0 < \gamma < 1$

Example Pole Balancing



Goal: Apply forces to cart so that pole does not fall.

Failure: . If pole falls past a given angle from the vertical

- Pole is reset to vertical position

Note: This is an episodic task

Rewards:

(i) $R = +1$ on every time step
where failure
did not occur
0 o.w.

[no discounting]

(ii) $R = -1$ on failure

0 o.w.

[Discounted return]

$$G_t = -\gamma^T \quad \text{if } 0 < \gamma < 1$$

Here $T \equiv$ time of failure

Policy: A map from states to probabilities
of selecting actions

- Under policy π , at time t ,

if $s_t = s$, then $A_t = a$
w.p. $\pi(a|s)$

Value function

$$V_{\pi}(s) \triangleq E_{\pi} [G_t | s_t = s]$$

↓
actions are
chosen according to π

$$= E_{\pi} \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} | s_t = s \right]$$

$\forall s \in \mathcal{S}$

- In Episodic tasks, we

allow $\gamma = 1$

- Action value function

$$q_{\pi}(s, a) \triangleq E_{\pi}[G_t \mid s_t = s, A_t = a]$$

$$= E_{\pi}\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid s_t = s, A_t = a\right]$$

Note:

$$V_{\pi}(s) = \sum_a \pi(a|s) q_{\pi}(s, a) \rightarrow (1)$$

Similarly,

$$q_{\pi}(s, a) = \tau(s, a)$$

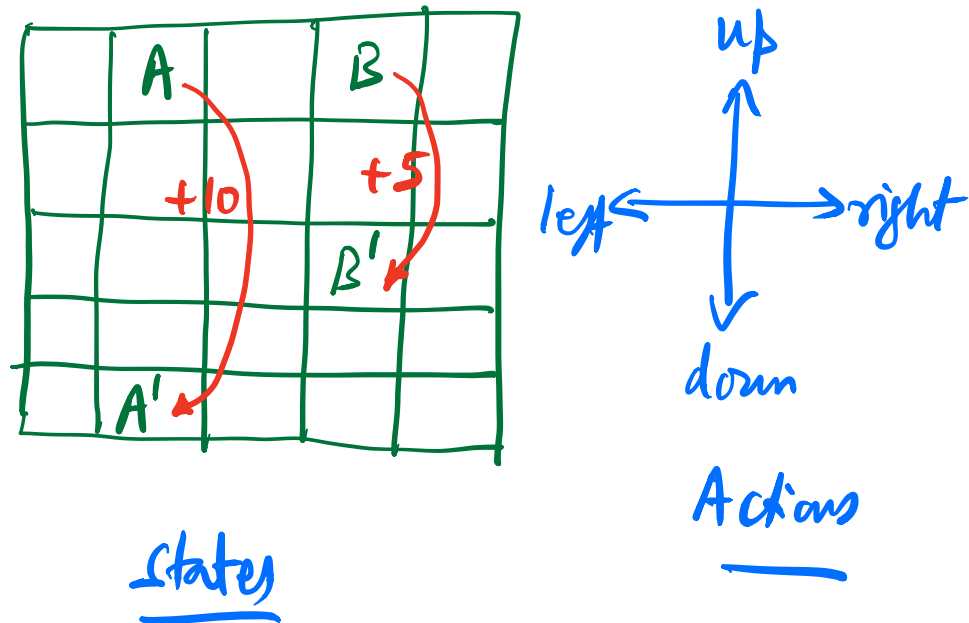
$$+ \gamma \sum_{\substack{s' \in \mathcal{S} \\ r \in \mathcal{R}}} p(s', r | s, a) U_{\pi}(s') \longrightarrow (2)$$

If we combine (1) & (2), we obtain

$$U_{\pi}(s) = \sum_a \pi(a|s) \left(r(s, a) + \gamma \sum_{\substack{s' \in \mathcal{S} \\ r \in \mathcal{R}}} p(s', r | s, a) U_{\pi}(s') \right) \longrightarrow (3)$$

Bellman Equation for policy π .

Example : Grid world



- Cells of grid : states of MDP
- Actions available : up, down, right, left
 - ↳ makes agent move one step in the suggested direction

- Actions taking agent out of grid leave position unchanged but give reward $= +1$
- Reward for other actions $= 0$.
- A & B are special states
 - All 4 actions in A lead to A' with reward $= +10$
 - All 4 actions in B lead to B' with reward $= +5$
- Consider the ~~equiprobable~~ policy

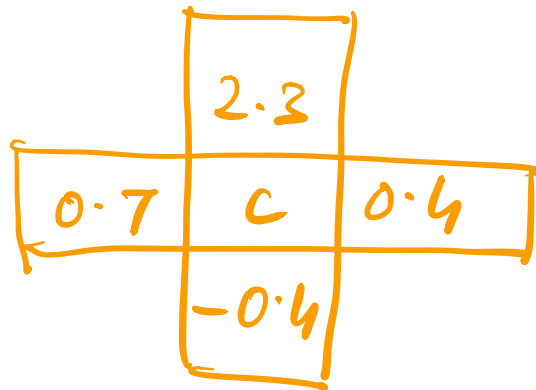
(π) - selects all 4 actions in each state with equal probability

• $\gamma = 0.9$ (discount factor)

• Value function V_π for policy π

3.3	8.8	4.4	5.2	1.5
1.5	3.0	2.3	1.9	0.5
0.1	0.7	0.7	0.4	-0.4
-1.0	-0.4	-0.4	-0.6	-1.2
-1.9	-1.3	-1.2	-1.4	-2.0

Consider state C



$$1. U_{\pi}(C) = \frac{1}{4} (0+0+0+0)$$

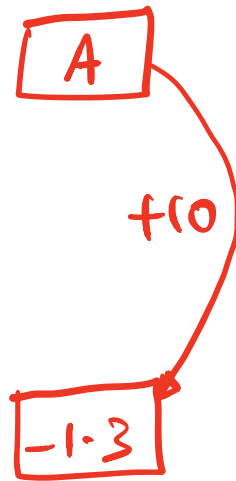
$$+ 0.9 \left(\frac{2.3 + \cancel{0.4} - \cancel{0.4} + 0.7}{4} \right)$$

$$= 0.9 \times \frac{2}{4} = 0.9 \times 0.75$$

$$\approx 0.675$$

$$\approx 0.7$$

2.



$$V_{\pi}(A) = \frac{1}{4} (10 + 10 + 10 + 10) + 0.9 * (-1.3)$$

$$= 10 - 1.3 * 0.9$$

$$= 10 - 1.17$$

$$= 8.83 \approx 8.8$$

3.

	-1.2
-1.4	Δ

$$\begin{aligned}
 V_{\pi}(\Delta) = & \frac{1}{4} * 0 + \frac{1}{4} * 0 \\
 & + \frac{1}{4} * (-1) + \frac{1}{4} * (-1) \\
 & + 0.9 * \frac{1}{4} \left(-1.2 - 1.4 \right. \\
 & \quad \left. + V_{\pi}(\Delta) \right. \\
 & \quad \left. + V_{\pi}(\Delta) \right)
 \end{aligned}$$

Thus

$$V_{\pi}(\Delta) = -\frac{1}{2} + 0.225 \left(-2.6 + 2 V_{\pi}(\Delta) \right)$$

$$= -0.5 - 2.6 \times 0.225 + 0.45 V_{\pi}(A)$$

$$\Rightarrow 0.55 V_{\pi}(A) = -0.5 - 0.625$$

$$= -1.125$$

$$\therefore V_{\pi}(A) \approx -2$$

Optimal policies and optimal value function

- An optimal policy is a policy that gives the largest value function across policies

$$\begin{aligned}
 \bullet \quad \text{Let } U_{\pi^*}(s) &= \max_{\pi} U_{\pi}(s) \\
 &\quad \forall s \in S \\
 &= U_{\pi^*}(s)
 \end{aligned}$$

Here $\pi_{\pi^*} \equiv \text{Optimal policy}$

Likewise, let

$$\begin{aligned}
 q_{\pi^*}(s, a) &= \max_{\pi} q_{\pi}(s, a) \\
 &\triangleq q_{\pi^*}(s, a)
 \end{aligned}$$

Here,

$$q_{\pi^*}(s, a) = E \left[R_{t+1} + \gamma U_{\pi^*}(s_{t+1}) \mid s_t = s, A_t = a \right]$$

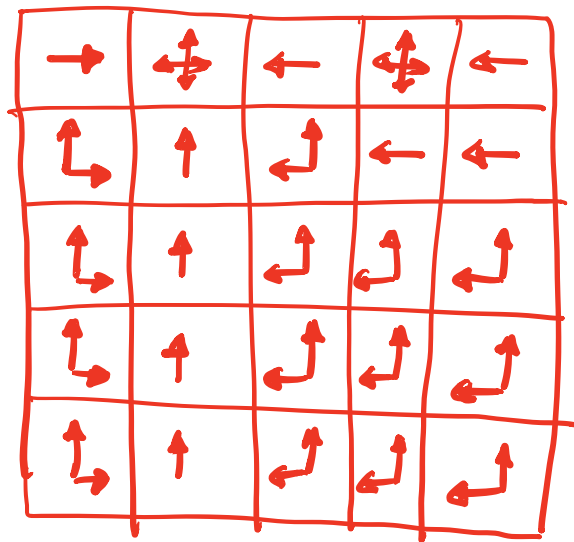
Bellman Equation

$$U_{*}(s) = \max_a \sum_{s', r} P(s', r | s, a) \left[r + \gamma U_{*}(s') \right]$$

On the grid world example,

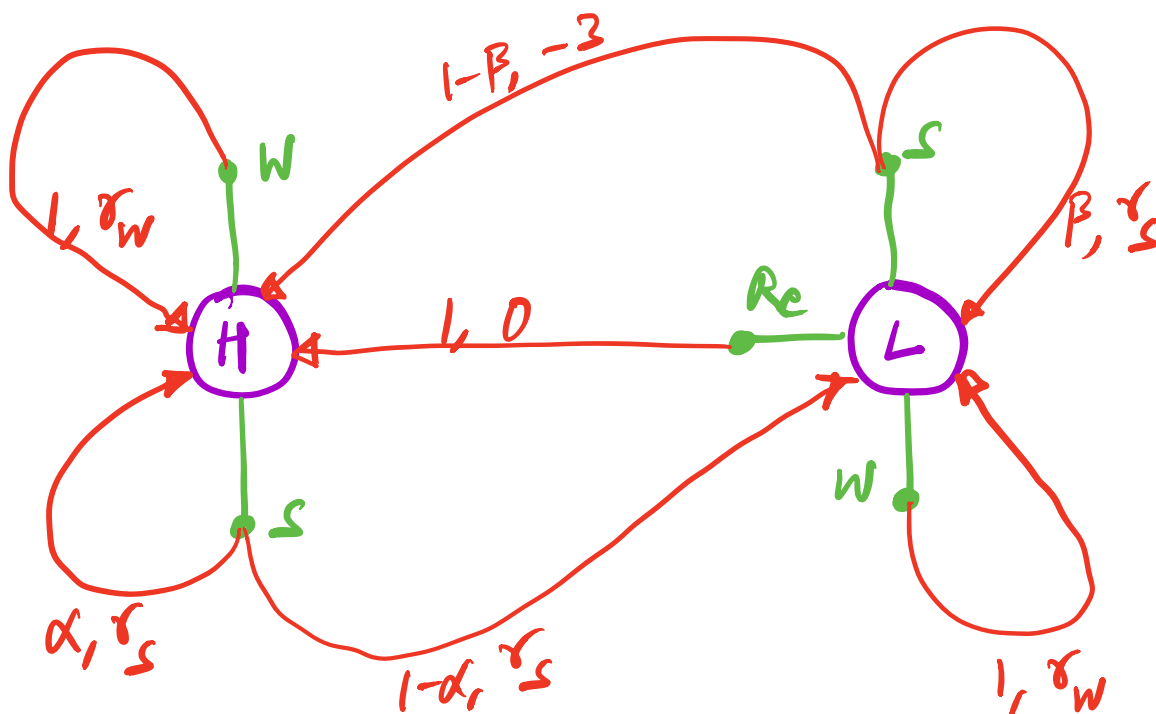
Optimal value function U_{*}

22.0	24.4	22.0	19.4	17.5
19.8	22.0	19.8	17.8	16.0
17.8	19.8	17.8	16.0	14.4
16.0	17.8	16.0	14.4	12.0
14.4	16.0	14.4	13.0	11.7



π^* (optimal policy)

Recall the recycling robot Example



Recycling Robot Example

$$\begin{aligned}
 v_*(H) = & \max \left\{ \alpha (r_S + \gamma v_*(H)) \right. \\
 & \quad \left. + (1-\alpha) (r_S + \gamma v_*(L)), \right. \\
 & \quad \left. 1 (r_H + \gamma v_*(H)) \right\}
 \end{aligned}$$

$$= \max \left\{ \sigma_L + \gamma \left[\alpha \underline{v}_*(H) + (1-\alpha) \underline{v}_*(L) \right], \right. \\ \left. \sigma_W + \gamma \underline{v}_*(H) \right\}$$

$$\underline{v}_*(L) =$$

$$\max \left\{ \beta \sigma_L + \gamma \beta \underline{v}_*(L) - \beta(1-\beta) + \gamma(1-\beta) \underline{v}_*(H), \right.$$

$$\sigma_W + \gamma \underline{v}_*(L),$$

$$0 + \gamma \underline{v}_*(H) \left. \right\}$$

$$= \max \left\{ \begin{aligned} &\beta r_L - \beta(1-\beta) \\ &\quad + \gamma \left[(1-\beta) U_*(H) \right. \\ &\quad \quad \left. + \beta U_*(L) \right] \\ &r_H + \gamma U_*(L), \\ &\gamma U_*(H) \end{aligned} \right\}$$

In next class, we shall
talk about numerical solution
approaches for solving Bellman
Equation.